Historical source

Encyclopaedia Britannica scan, source row 1032. Parse 3.5 extracted the text. Super 3.5 VL corrected it from an enlarged crop of each block. It can still contain recognition errors. Page headers, footers, and page numbers are removed.

Untitled continuation (E0089)

Continues an entry from an earlier page.

tery of a man, and that the inward area of the left ventricle of his heart is equal to fifteen square inches; these multiplied into 7+1/2 feet, give 1350 cubic inches of blood, which prefixes on that ventricle, when it first begins to contract, a weight equal to 15.5 pounds.

What the doctor thus calculates, from supposition, with regard to mankind, he actually experimented upon horses, dogs, fallow-does, &c. by fixing tubes, in orifices opened in their veins and arteries; by observing into federal heights, or which the blood rode in these tubes, as they lay on the ground, and by measuring the capacities of the ventricles of the heart, and orifice of the arteries. And, that the reader may the more readily compare the said estimates together, he has given a table of them, ranged in the following order.

\begin{tabular}{ccccccccccc} **The feeral animals.** & **Weight of each.** & **Height of the blood in the tube from the gui- urine.** & **Capacity of the left ventricle of the heart.** & **Area of the orifice of the aorta.** & **Velocity of the blood in the sorta.** & **Welocity of the blood in the sorta.** & **Flows.** & **How much in a minute** & **Weight of the blood fur- flined by the left ventricle contracting.** & **Area of transferfe fiction of defending aorta.** & **Area of the transferfe fiction of defending aorta.**\\ & Pounds, Ounces. & Inches. & Retelaches. & Cubicinches & Square inches. & Pete andiae circa in minute. & Minutes. & Pounds. & Pounds. & No of pullies in a minute. & Square inches.\\ Man & 160 & On drain- ing. & 7 6 & 1.659 & 3.318 & 0.4187 & 56.55 & 34.18 & 34.38 & 51.5 & \\ Horfe 1th. 2d. 3d. & 12 5 & 9 8 & 8 9 & 10 2.5 & 1.036 & 86.85 & 60 88 & 13.75 & 113.22 & 86 38 & 0.369\\ Ox & 835 1600 & 12 5 & 9 6 & 10 1.539 & 86.95 & 76.95 & 18.14 & 113.22 & 86 38 & 0.677 0.912 & 0.84\\ Sheep Doe & 91 & 5 & 9 6 & 1.85 & 0.172 0.476 & 174.5 & 20 & 4.593 & 36.56 & 50 0.984 & 0.07 0.012\\ Dogs ift. & 52 24 & 5 6 & 6 8 8 & 1.172 & 0.196 & 144.77 & 11.9 & 4.34 & 33.61 & 97 & 0.106 0.034\\ 2d. 3d. 3d. & 18 5 & 5 7 & 8 8 & 0.185 & 0.118 & 130 & 7.8 & 2.3 & 19.8 & 11.1 & 0.07 0.022 0.009\\ 4th. & 12 8 & 4 & 8 3 & 0.633 & 0.101 & 120 & 6.7 & 1.85 & 11.1 & 0.061 & 0.015 0.007\\ \end{tabular}

HEAT (E0090)

HEAT, in physiology, one of the secondary qualities of bodies, produced by fire, and opposed to cold.

Under the article fire, we considered the fun as the principal source of heat upon the earth's surface, and the confines of the earth and atmosphere: without this, all the bodies upon our globe would doubtless grow rigid, lifeless, and fixed. It is this that flies within them, as the main spring of their actions. Hence vegetation and animalization are evidently promoted; and hence the ocean and the atmosphere continue in a fluid state.

Heat in us is properly a sensation, excited by the action of fire; or it is the effect of fire on our organs of feeling. Hence it follows, that what we call heat is a particular idea or modification of our own mind, and not any thing exifling in that form in the body that occasions it. Heat, fays Mr Locke, is no more in the fire that burns the finger, than pain is in the needle that pricks it. In effect, heat in the body that

gives it, is only motion ; and in the mind, only a par- ticular idea.

Heat in the hot body, according to 'S Gravefande, is an agitation of the parts of the body, made by means of the fire contained in it : by such an agitation a motion is produced in our bodies, which excites the idea of heat in our mind ; so that heat in respect of us is nothing but that idea, and in the hot body nothing but motion. If such motion expel the fire in right lines, it gives us the idea of light ; if in a various and irregular motion, only heat.

Heat, with respect to our sensations, or the effect produced on us by a hot body, is estimated by its relation to the organ of feeling; no object appearing to be hot, unless its heat exceed that of our body. Whence the same thing to different persons, or at different times to the same person, shall appear both hot and cold. The degree of heat is measured by the expansion of the air, or spirit in the thermometer.

Source record

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